

LG Patrols Effort Workflow

This guide walks you through loading, configuring, and running the LG Patrols Effort Workflow, which generates a patrol effort analysis report for Lion Guardians rangers in the Amboseli ecosystem sourced from EarthRanger.

What it produces

The workflow produces, for each patrol group:

- A **Patrol Events map** (scatter plot of events, coloured by event type)
 - A **Patrol Trajectories map** (tracks in a single fixed colour)
 - A **Linear Time Density map** (patrol coverage heat map)
 - A **time series bar chart** and **pie chart** of patrol events
 - **Summary CSV tables** — per guardian, event type, and monthly breakdown
 - A **Word document report** (`.docx`) with a cover page and one section per group
 - An **interactive widget dashboard**
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Requirements

- Access to an **EarthRanger** instance with a configured data source
- The patrol types and event types you want to analyse (leave blank to include all)

The two study-area boundaries (Conservancies and Group Ranch Boundaries) are fetched live from EarthRanger — no local files are required. Only the Word report templates and the organisation logo are downloaded automatically from Dropbox.

1. Load the Workflow

In the workflow runner, go to **Workflow Templates** and click **Add Workflow Template**. Paste this repository's URL into the **Github Link** field, then click **Add Template**:

Once added, it appears in the **Workflow Templates** list as **lg-patrols-effort**. Click it to open the workflow configuration form.

The card may show **Initializing...** briefly while the environment is set up.

You'll also need an **EarthRanger data source connection** configured beforehand (via **Data Sources** in the runner) — patrol tracks and events are pulled live from EarthRanger, so a connection is required before this workflow can run.

2. Configure the Workflow

Workflow Details and Time Range

Field	Description
Workflow Name	A short name to identify this run
Workflow Description	Optional notes about the run (e.g. date range or patrol group)

Time range

Field	Description
Timezone	Select the local timezone (e.g. <code>Africa/Nairobi UTC+03:00</code>)
Since	Start date and time of the analysis period
Until	End date and time of the analysis period

All patrol tracks, events, and metrics are computed within this window.

Groupers

Groupers control how the workflow partitions data for per-group outputs. **Left blank by default** — all data appears in a single combined view. Click **Add** to add a grouper:

Grouper	Effect
Patrol Type	One map, metrics, and report section per patrol type

Group	Effect
Patrol Serial Number	One output per patrol serial
Patrol Status	One output per patrol status (e.g. done)
Patrol Subject	One output per guardian ranger

Connect to EarthRanger

Select the EarthRanger data source configured for this environment from the **Connect to EarthRanger** dropdown. Patrol tracks, patrol events, and the two study-area boundary layers (Conservancies and Group Ranch Boundaries) are all fetched live from this connection — there is no offline/local-file mode for this workflow.

Patrol and Patrol Event Parameters

Field	Description
Patrol Types	Filter to specific patrol types — defaults to <code>routine_patrol</code> only
Event Types	Filter to specific event types (leave empty to include all)
Include Events Without a Geometry	Check to include events with no point or polygon location — excluded by default

Expand **Advanced Configurations** to access additional query parameters, including patrol status (defaults to `done` patrols only) and date-range overlap behaviour.

Basemap Layers

Two stacked ArcGIS tile layers form the background of every map. Pre-filled with sensible defaults, but the URL, opacity, and max zoom of each layer are editable.

Layer	Default Opacity	Max Zoom
ESRI World Hillshade	<code>1.0</code>	<code>20</code>
ESRI World Street Map	<code>0.15</code>	<code>20</code>

Trajectory Segment Filter (*Advanced Configurations*)

These parameters remove GPS noise and biologically unrealistic movements before trajectory analysis. Patrol tracks are always rendered in a single fixed colour — there is no user-selectable colour category for this map.

Field	Default	Description
Minimum Segment Length (m)	10	Discard segments shorter than this distance
Maximum Segment Length (m)	100000	Discard segments longer than this distance
Minimum Segment Duration (s)	10	Discard segments shorter than this duration
Maximum Segment Duration (s)	21600	Discard segments longer than this duration (6 hours)
Minimum Segment Speed (km/h)	1	Discard segments below this average speed
Maximum Segment Speed (km/h)	7	Discard segments above this average speed

A daytime filter also restricts patrol fixes to between **06:00 and 19:00** local time, excluding night-time GPS drift.

Filter Patrol Events *(Advanced Configurations)*

Restricts events to a bounding box, pre-filled by default to the Amboseli ecosystem extent. Clear it to include all events within the time range regardless of location.

Time Series Bar Chart

Field	Description
Time Interval	The time bucket used to group events on the x-axis (e.g. day, week, month)

Linear Time Density Meshgrid *(Advanced Configurations)*

Controls the resolution of the patrol coverage raster. Leave at defaults for most analyses.

3. Run the Workflow

Once all parameters are configured, click **Submit**. The runner will:

1. Pull patrol tracks and events from EarthRanger for the specified time range.
2. Fetch the Conservancies and Group Ranch Boundaries study-area layers live from EarthRanger.
3. Convert observations to relocations and build trajectory segments.
4. Generate the Patrol Events, Patrol Trajectories, and Linear Time Density maps.
5. Compute per-guardian, event type, and monthly summary tables.
6. Render the time series bar chart and pie chart.
7. Assemble the Word report (cover page + per-group sections) and the dashboard.

8. Save all outputs to the directory specified by `ECOSCOPE_WORKFLOWS_RESULTS` .

Output Files

All outputs are written to `$ECOSCOPE_WORKFLOWS_RESULTS/` :

File	Description
<code>events.geoparquet</code>	Patrol events with display names and timezone-converted timestamps
<code>trajectories.geoparquet</code>	Patrol segments with speed and distance
<code><group>_events.html</code>	Interactive patrol events map per group
<code><group>_patrol_trajectories.html</code>	Interactive patrol trajectories map per group
<code><group>_time_density.html</code>	Interactive linear time density map per group
<code><group>_events.png</code>	Screenshot of the patrol events map (2× resolution)
<code><group>_patrol_trajectories.png</code>	Screenshot of the patrol trajectories map (2× resolution)
<code><group>_time_density.png</code>	Screenshot of the time density map (2× resolution)
<code><group>_patrols_pie_chart.png</code>	Screenshot of the event type pie chart (2× resolution)
<code><group>_patrol_events_time_series_bar_chart.png</code>	Screenshot of the events time series bar chart (2× resolution)
<code><group>_guardian_patrol.csv</code>	Per-guardian patrol effort (patrols, distance, time)
<code><group>_guardian_events.csv</code>	Per-guardian event count
<code><group>_pivot_guardian_events.csv</code>	Guardian × event type pivot table
<code><group>_monthly_patrol_efforts.csv</code>	Monthly patrol effort summary
<code><group>_event_types.csv</code>	Per-event-type count
<code>cover_page.docx</code>	Rendered report cover page
<code><group>.docx</code>	Per-grouper report section
<code>overall_report.docx</code>	Final combined Word report

A patrol-type summary is computed internally (`summarized_patrol_types`) but is not currently persisted to a CSV or wired into the report/dashboard.